

33.2 A Fully Integrated Analog ReRAM Based 78.4TOPS/W Compute-In-Memory Chip with Fully Parallel MAC Computing

Qi Liu¹, Bin Gao¹, Peng Yao¹, Dong Wu¹, Junren Chen¹, Yachuan Pang¹, Wenqiang Zhang¹, Yan Liao¹, Cheng-Xin Xue², Wei-Hao Chen², Jianshi Tang¹, Yu Wang¹, Meng-Fan Chang², He Qian¹, Huaqiang Wu¹

¹Tsinghua University, Beijing, China

²National Tsing Hua University, Hsinchu, Taiwan

Non-volatile memory (NVM) based computing-in-memory (CIM) shows significant advantages in handling deep learning tasks for artificial intelligence (AI) applications. To overcome the decreasing cost effectiveness of transistor scaling and the intrinsic inefficiency of data-shuttling in the von-Neumann architecture, CIM is proposed to realize high-speed and low-power system with parallel multiplication accumulation (MAC) computing [1][2]. However, current demonstrations are mainly based on single macro and present limited computing parallelism. Realizing a fully-integrated CIM chip with a complete neural network model is still missing. The major challenges lie in: (1) The IR drop and transient errors when carrying out MAC operations in non-volatile memory arrays decrease the computing accuracy and further limit the parallelism; (2) The inefficiency of the interface blocks between different arrays due to the power overhead of the A/D and D/A converters (shown in Fig. 33.2.1). To address these challenges, this work proposes: (1) A sign-weighted 2T2R (SW-2T2R) array to reduce IR drop by decreasing the accumulative SL current (ISL), and eventually boost the computing parallelism; (2) a low-power interface design with resolution-adjustable LPAR-ADC to realize flexible tradeoff between system accuracy and power consumption. In this manner, this work implements a fully-integrated 784-100-10 MLP model on an integrated CIM chip with 158.8Kb analog ReRAMs. This chip realizes high recognition accuracy (94.4%) on MNIST database, high inference speed (77 μ s/Image), and 78.4 TOPS/W peak energy efficiency. The CMOS circuits are fabricated in a 130nm process.

Figure 33.2.2 presents the algorithm, structure and work flow diagram regarding the proposed ReRAM-based CIM chip. This work realizes a two-layer perceptron model, which consists of two fully-connected weight arrays and three neural layers. Accordingly, the chip structure is composed of a SW-2T2R array, a 1T1R array, input/output buffers, LPAR-ADCs, etc. In the SW-2T2R array, the positive weight and negative weight in a differential device pair are connected on the same output column, which is different from Ref. [2] or [3]. An x-bit signed weight (1-bit sign, x-1-bit data) is stored in a SW-2T2R cell. During n parallel MAC operations, n LPAR-ADCs clamp the SLs to voltage (V_{SL_CLP}) and convert SL currents to digital outputs. The SL current is the accumulation result of all the SW-2T2R cell currents on a same column. Each MAC operation evaluates the product of one m-dimensional 1bit-input vector and an x-bit sign-weight vector. The output of LPAR-ADC is stored in registers and sampled simultaneously to the next ReRAM array as the input data. The resolution of LPAR-ADC is adjustable by changing the sampling clock frequency. The flexible configuration of the interface block helps to achieve the balance between the system accuracy and power consumption. The output of the second ReRAM array i.e. MAC2-OUT, is sampled by counters and stored in output buffers (shown in Fig. 33.2.2). If the resolution of the 1st-stage ADC is configured as N1 bit, it will generate 2^{N1} pulses. Similarly, if the resolution of the 2nd-stage ADC is set as N2 bit, for each pulse of the 1st-stage output, the 2nd-stage ADC generates 2^{N2} pulses. Thus, the inference time of one image will last for $2^{(N1+N2)}$ cycles at least. A higher ADC resolution could lead to better system recognition accuracy, while consuming more energy and latency.

Figure 33.2.3 presents the structure, operating timing diagram and truth table of SW-2T2R array. In a SW-2T2R cell, two ReRAMs represent positive and negative weights by utilizing opposite voltage polarity during inference stage. If $V_{SL} = V_{CLP}$, $V_{BLP} = V_{CLP} - V_{READ}$, $V_{BLN} = V_{CLP} + V_{READ}$, G_{POS} and G_{NEG} would represent the positive and negative weight respectively. Eventually, the equivalent weight of this SW-2T2R cell is $W_{CELL} = (G_{POS} - G_{NEG})$, which could be positive, negative or zero. The SL current accounted for this weight pair is equal to the differential currents flowing through the positive cell and the negative cell, respectively. This current is proportional to W_{CELL} according to $I_{CELL} = V_{READ} * W_{CELL}$. The 2T2R structure is designed to improve CIM accuracy by reducing the IR drop from two aspects: (a) if $G_{POS} = G_{NEG}$, I_{CELL} can be reduced to zero; (b) The current through the positive weight and negative weight on the same column can be cancelled out locally. ReRAM precision determines the weight precision of a SW-2T2R cell, shown in the truth tables of Fig. 33.2.2. The ReRAM-based weight presentation is defined according to the device intermediate states. If single ReRAM device works as a 1-bit (2 device levels) or 2-bit (4 device levels), the weight precision of a SW-2T2R

cell is signed quasi-2-bit (3-level) or signed quasi-3-bit (7-level) accordingly. According to the off-chip test on the same ReRAM stacks, the device conductance could be tuned continuously. However, the on-chip ADC resolution limits that the on-chip ReRAM conductance could be quantified with 256 states at most.

Figure 33.2.4 shows the structure and timing diagram of LPAR-ADC. LPAR-ADC is composed of three sub-modules: an integrator, a comparator and a segmented-capacitor DAC (SC-DAC). The integrator consists of an operational amplifier (OPA) and an integrating capacitor. The integrator clamps the SL to V_{CLP} and converts the SL current to an analog voltage signal. The SC-DAC generates a ramp voltage signal from V_{CLP} to V_{DD} . The comparator is used to compare the ramp voltage signal and the integrated voltage signal. The ADC workflow includes three phases: (1) PH1: Reset ADC via keeping the RST_integ/EN_integ switch 'ON', and the EN_DAC/EN_comp switch 'OFF'. In this phase, SL and OUT_integ are clamped to V_{CLP} , and V_{RAMP} remains at the initial voltage, i.e., V_{CLP} . (2) PH2: Sample I_{SL} . In this phase, the RST_integ switch is cut off and Cinteg is accessed to the SL current. Charges from the SL and OPA are accumulated on the capacitor. The voltage of OUT_integ is then changed accordingly. EN_comp is enabled to pre-charge the comparator. (3) PH3: MAC-OUT. In this phase, the EN_integ switches off to cut off the SL current read-path. Meanwhile, the voltage of OUT_integ is maintained. SC-DAC starts to count and generates a ramp voltage signal. The comparator compares the output of the integrator and SC-DAC, and generates a spike pulse. As illustrated in Fig. 33.2.2, the power consumption of LPAR-ADC is controlled by the reference current source of the integrator and comparator, and the ADC resolution is configured by setting the frequency of the sampling clock. It is worth mentioning that integration and quantization method could filter out the current overshoot and fluctuation by averaging the accumulative ISL over the integrating period. The quantized output minimizes the transient errors.

Figure 33.2.5 shows the experimental results on access time, power consumption, accuracy and speed during inference on the MNIST dataset. In the case of $V_{DD} = 4.2V$ and $V_{READ} = 0.2V$, MAC-OUT access time is 51.1ns. The test result shows that the power consumption of SW-2T2R CIM chip is 1.9 $\times$ lower than that of a 1T1R version. All the data are obtained based on a same CIM chip with different weight structure. In addition, the test results show that the recognition accuracy of MNIST dataset is positively correlated with the ADC resolution of both stages, while the inference speed is positively correlated with the resolution of the 1st stage ADC. When the resolution of the 1st/2nd stage ADC is configured as 2bit/8bit, the recognition accuracy is ~92% and the inference speed is 77 μ s/Image.

Figure 33.2.6 shows the test system and software interface when testing the CIM chip. The test system includes a FPGA board, a test chip board and a host computer. The FPGA is used to move data and commands between the host computer and the CIM chip. The simulation result of the 784-100-10 fully-connected NN shows that the SW-2T2R structure can effectively reduce accuracy loss due to the IR drop. Using 3-bit signed weight, the test result reaches an accuracy of 93.4%, which is ~2% lower than the simulation result. The comparison results between this work and prior works are summarized. This work has achieved better performance under less advanced technology in terms of peak energy efficiency, MNIST dataset recognition accuracy, ADC resolution and the inference speed.

Figure 33.2.7 shows the die photomicrograph, the layout of a SW-2T2R cell, and a specific feature table. In summary, this work implements a 158.8Kb ReRAM CIM chip in 130nm CMOS process. For the first time, a CIM chip is fully integrated for a complete multi-layer NN model, and recognizes the MNIST images at a high speed of 77 μ s/Image, 78.4 TOPS/W peak energy efficiency and 94.4% test accuracy.

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- [2] R. Mochida et al., "A 4M Synapses Integrated Analog ReRAM Based 66.5 TOPS/W Neural-Network Processor with Cell Current Controlled Writing and Flexible Network Architecture," *VLSI*, pp. 175-176, 2018.
- [3] C.-X. Xue et al., "A 1Mb Multibit ReRAM Computing-In-Memory Macro with 14.6ns Parallel MAC Computing Time for CNN Based AI Edge Processors," *ISSCC*, pp. 388-390, Feb. 2019.

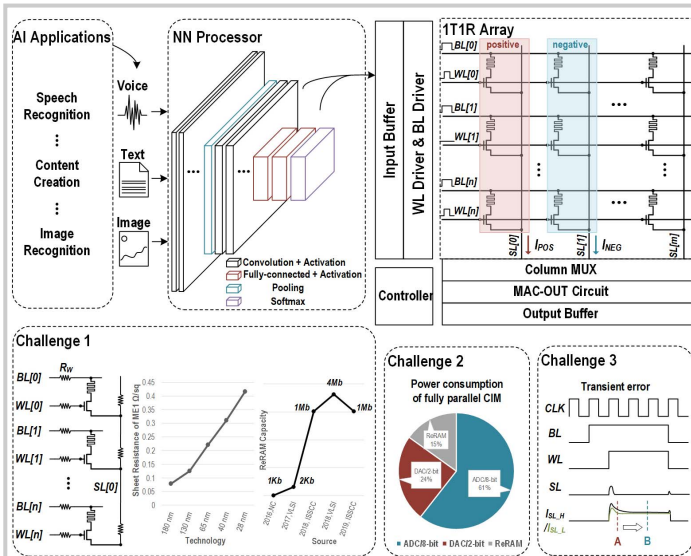


Figure 33.2.1: Computing-in-memory for AI applications.

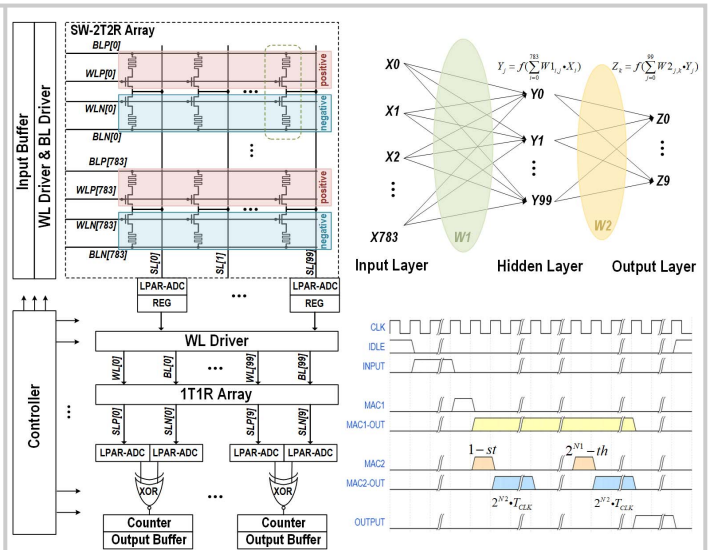


Figure 33.2.2: Proposed fully-integrated computing-in-memory chip.

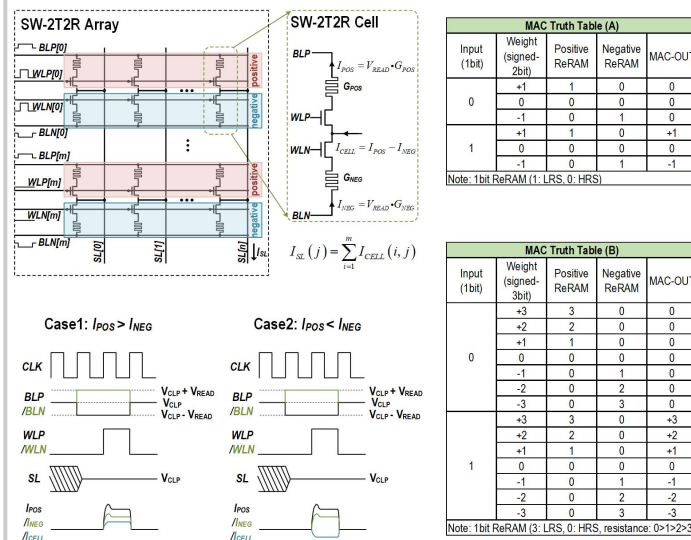


Figure 33.2.3: Signed-weighted 2T2R array.

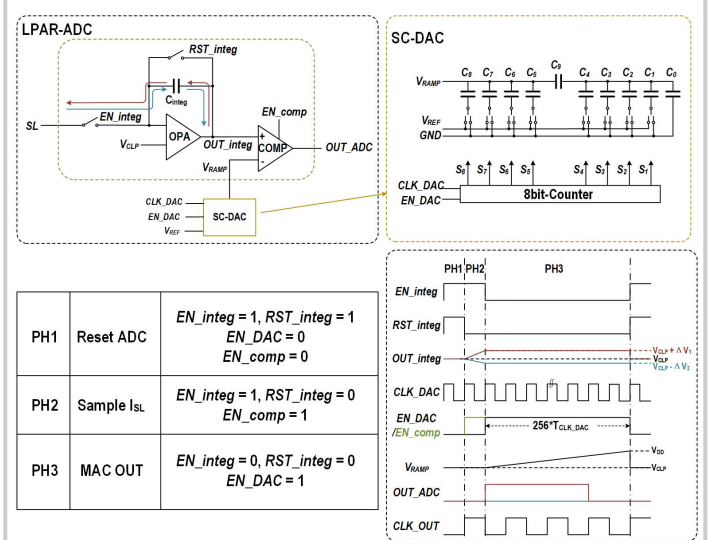


Figure 33.2.4: Structure and timing diagram of LPAR-ADC.

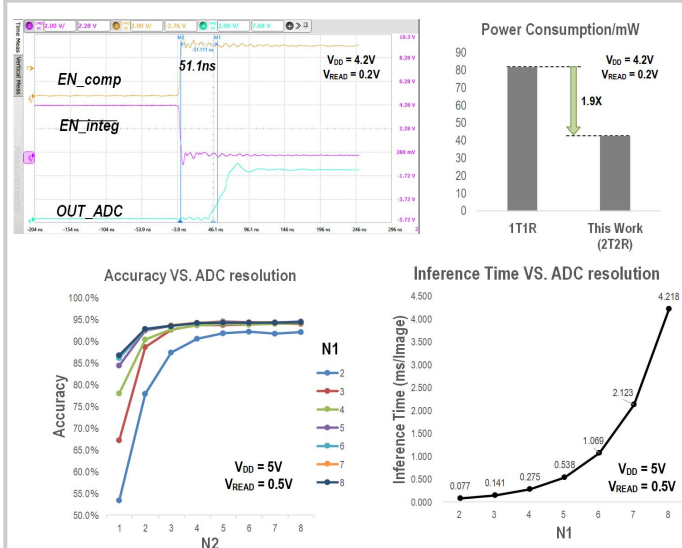


Figure 33.2.5: Measurement results during inference on MNIST dataset.

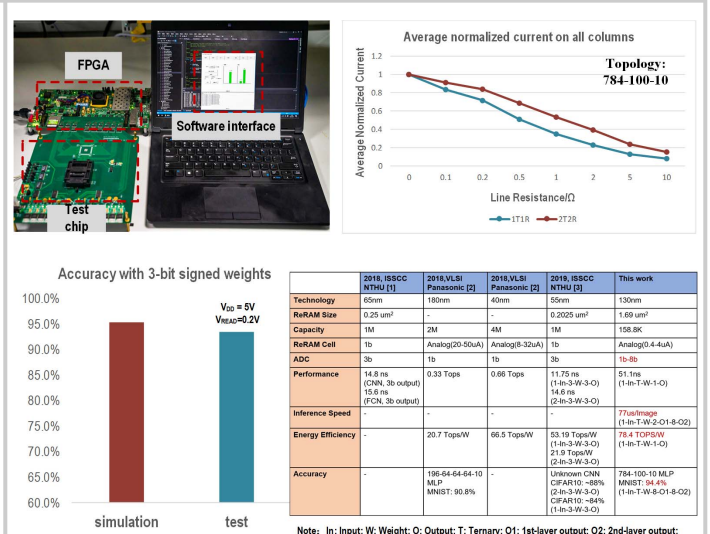


Figure 33.2.6: Test system and comparison table.

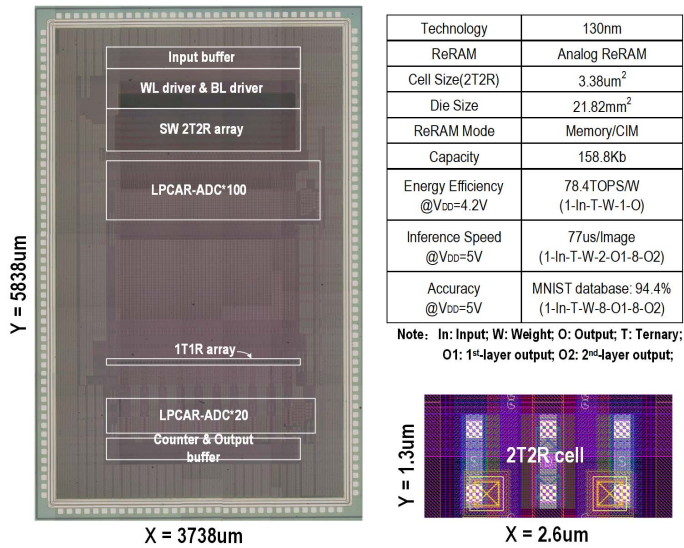


Figure 33.2.7: Die micrograph with SW-2T2R layout and feature table.

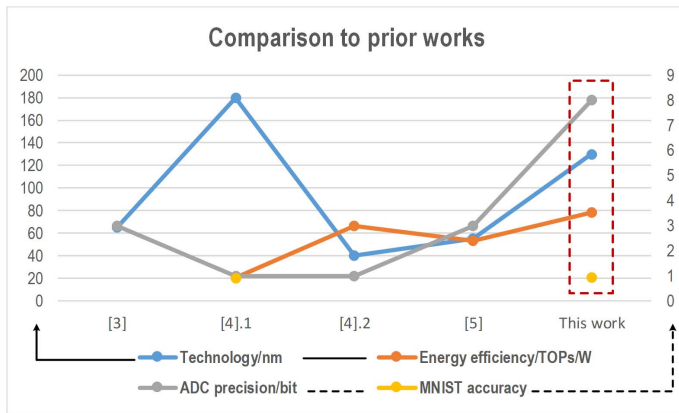


Figure 33.2.S2: Comparison diagram between this work and prior works.

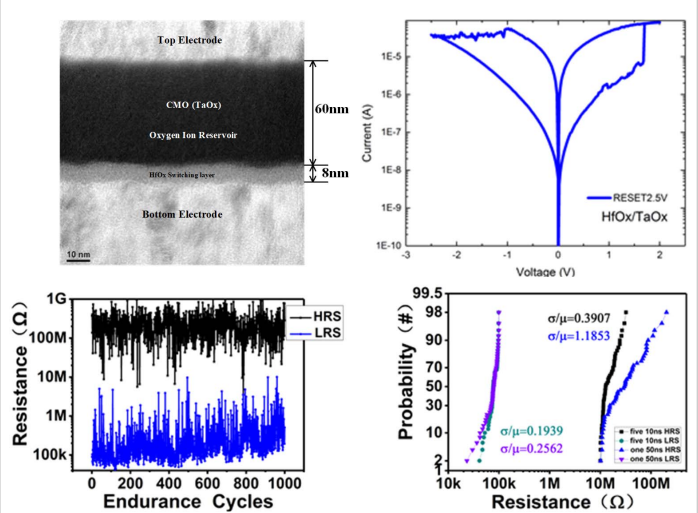


Figure 33.2.S1: Cross-sectional TEM image and device characteristics of ReRAM cells.

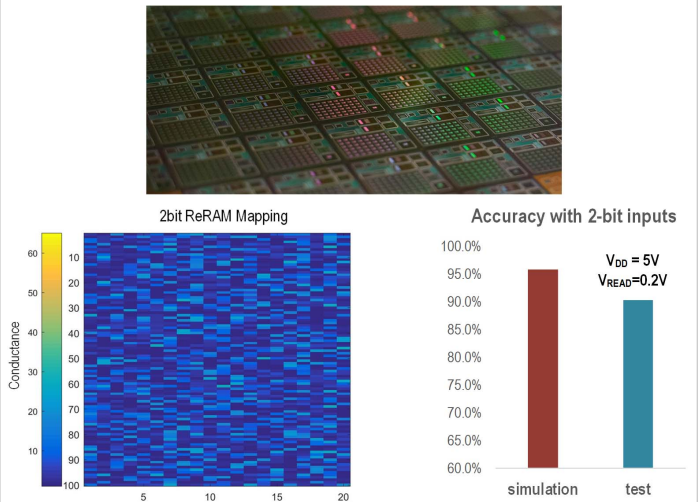


Figure 33.2.S3: Wafer micrograph, 2bit-ReRAM mapping and 2bit-input accuracy.